Statistical & Numerical Methods HW3

Daniel Haim Breger, 316136944

Technion

(Date: June 22, 2026)

1 Measurements in the Lab

Loading the data in python and fitting both a lorentzian and gaussian to the measurements, then inserting calculating the likelihood function using

$$L = \prod_i f(x_i) \tag{1}$$

we find

$$L(Gaussian) = 2 \times 10^{-118} \qquad L(Lorentzian) = 1 \times 10^{-104} \tag{2}$$

therefore the Lorentzian is a better fit, and is more likely by $\sim 5 \times 10^{13}$ times.

2 Moments of a distribution

2.1 Moments

Throwing the distribution into python and using sympy gives the moments:

$$\langle X\rangle = \begin{cases} \frac{ab\Gamma\left(1-\frac{1}{a}\right)\Gamma\left(p+1+\frac{1}{a}\right)}{(ap+1)\Gamma(p)} & \text{for } \frac{1}{a} < 1 \\ ab^a p \int\limits_0^\infty x^{ap}\left(b^a + x^a\right)^{-p-1}\, dx & \text{otherwise} \end{cases} \tag{3}$$

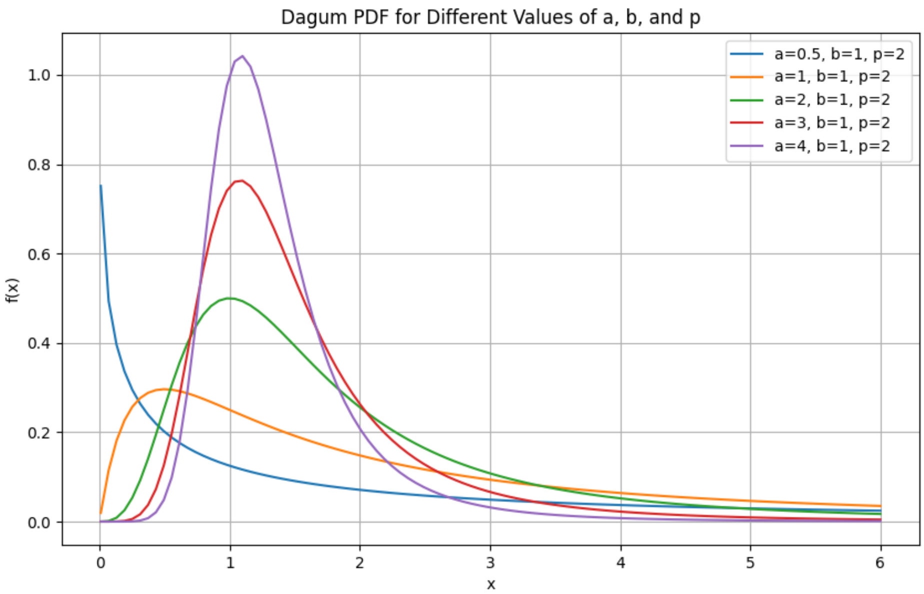
$$\langle X^2\rangle = \begin{cases} \frac{ab^2\Gamma\left(1-\frac{2}{a}\right)\Gamma\left(p+1+\frac{2}{a}\right)}{(ap+2)\Gamma(p)} & \text{for } \frac{1}{a} < \frac{1}{2} \\ ab^a p \int\limits_0^\infty x^{ap+1}\left(b^a + x^a\right)^{-p-1}\, dx & \text{otherwise} \end{cases} \tag{4}$$

$$\langle X^3\rangle = \begin{cases} \frac{ab^3\Gamma\left(1-\frac{3}{a}\right)\Gamma\left(p+1+\frac{3}{a}\right)}{(ap+3)\Gamma(p)} & \text{for } \frac{1}{a} < \frac{1}{3} \\ ab^a p \int\limits_0^\infty x^{ap+2}\left(b^a + x^a\right)^{-p-1}\, dx & \text{otherwise} \end{cases} \tag{5}$$

$$\langle X^4\rangle = \begin{cases} \frac{ab^4\Gamma\left(1-\frac{4}{a}\right)\Gamma\left(p+1+\frac{4}{a}\right)}{(ap+4)\Gamma(p)} & \text{for } \frac{1}{a} < \frac{1}{4} \\ ab^a p \int\limits_0^\infty x^{ap+3}\left(b^a + x^a\right)^{-p-1}\, dx & \text{otherwise} \end{cases} \tag{6}$$

2.2 Set of values

The PDF is

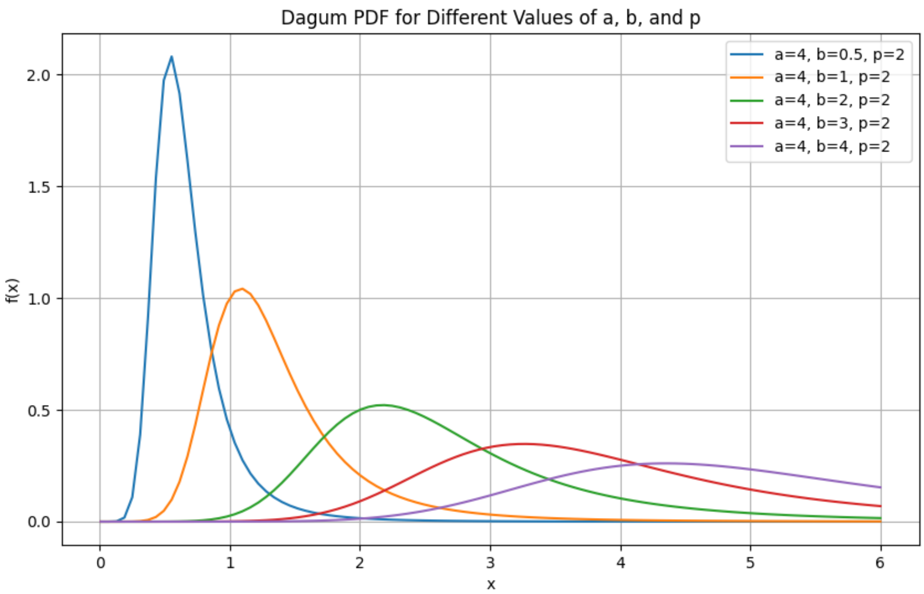


and the moments are

Moment	a=0.5	a=1	a=2	a=3	a=4
1	-	-	$\frac{3\pi}{4}$	1.61	1.39
2	-	-	-	4.03	$\frac{3\pi}{4}$
3	-	-	-	-	5.83

2.3 other values

The PDF is

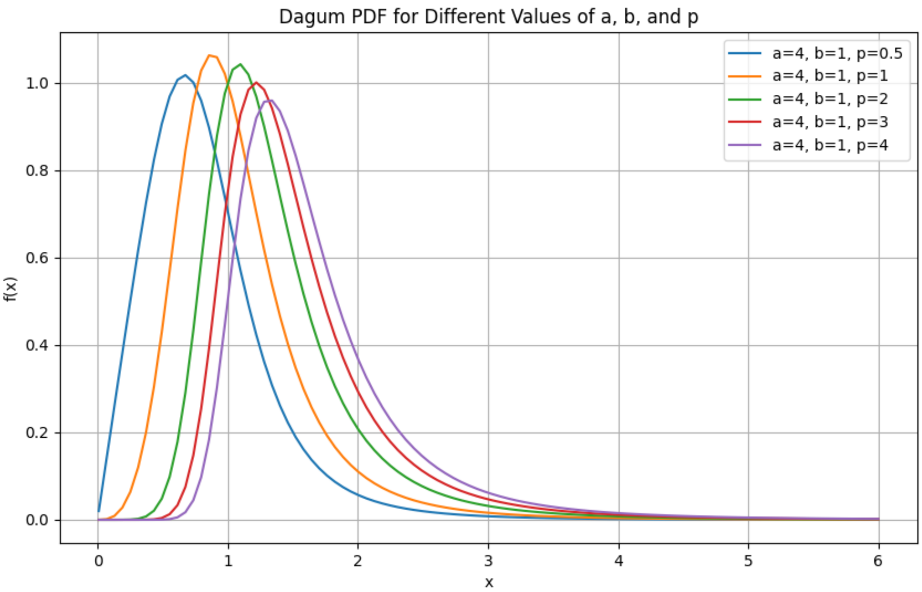


and the moments are

Moment	b=0.5	b=1	b=2	b=3	b=4
1	0.694	1.39	2.78	4.17	5.55
2	$\frac{3\pi}{16}$	$\frac{3\pi}{4}$	3π	$\frac{27\pi}{4}$	12π
3	0.729	5.83	46.6	157	373

2.4 yet more values

The PDF is



and the moments are

Moment	p=0.5	p=1	p=2	p=3	p=4
1	0.847	1.11	1.39	1.56	1.69
2	1	$\pi/2$	$\frac{3\pi}{4}$	$\frac{15\pi}{16}$	$\frac{35\pi}{32}$
3	1.85	3.33	5.83	8.02	10

2.5 fourth moment

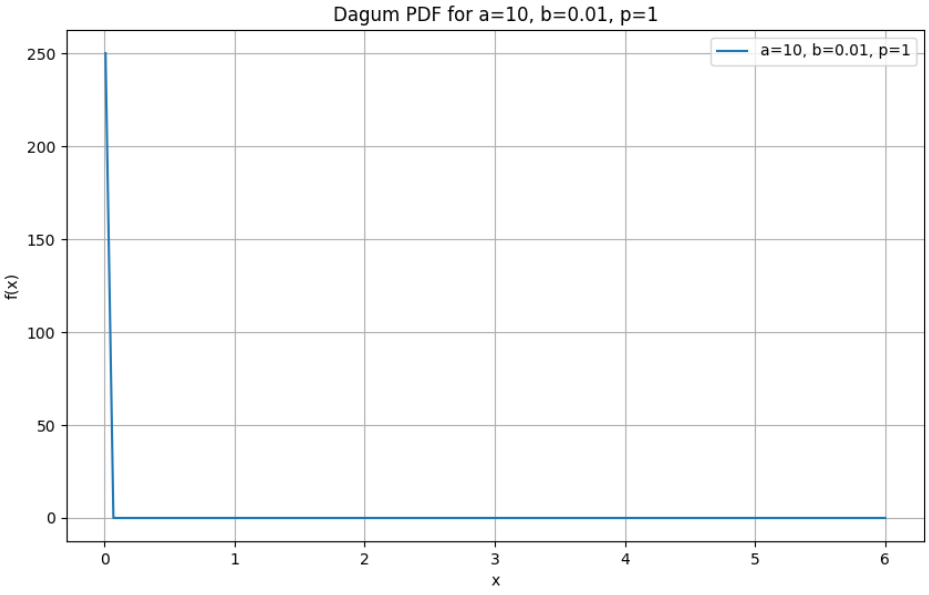
By definition the fourth moment is

$$\langle X^4 \rangle = \int_0^\infty x^4 f(x) \tag{7}$$

which for the dagum distribution is always positive, so no it's not possible to get negative fourth moments. To get a small fourth moment we want a distribution that decays to 0 very fast. Some trial and error gave me

$$a = 10 \qquad b = 0.01 \qquad p = 1 \tag{8}$$

visually



for which

$$\langle X^4 \rangle \approx 7\Gamma(3/5)\Gamma(12/5) \approx 1.3 \times 10^{-8} \tag{9}$$

3 Radioactive Decay

It's known radioactive decay follows an exponential decay of the form

$$N(t) = N_0 e^{-\lambda t} \tag{10}$$

since we were asked to find the chi-squared, we look at the definition

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - f(x_i, a)}{\sigma_i} \right)^2 = \sum_{i=1}^{N} \left(\frac{N_i - N_0 e^{-\lambda t}}{\sigma_i} \right)^2 \tag{11}$$

but σ_i is a constant and the same for all measurements so

$$\chi^2 = \frac{1}{\sigma^2} \sum_{i=1}^{N} \left(N_i - N_0 e^{-\lambda t} \right)^2 \tag{12}$$

We want to find the halflife for which chi-square is minimal so

$$\frac{\partial \chi^2}{\partial \lambda} = \frac{2}{\sigma^2} \sum_{i=1}^{N} \left(N_i - N_0 e^{-\lambda t} \right) N_0 t e^{-\lambda t} = \frac{2N_0}{\sigma^2} \sum_{i=1}^{N} t \left(N_i - N_0 e^{-\lambda t} \right) e^{-\lambda t} = 0 \tag{13}$$

plugging this into python we get

$$\lambda \approx 0.66 \tag{14}$$

so the halflife time is

$$\frac{1}{2} = e^{-\frac{2t}{3}} \implies t = -\frac{3}{2} \ln(0.5) \approx 1.04\text{h} \tag{15}$$

Now to calculate the error on this we notice that for each measurement the halflife time as calculated depends on the counts in the specific measurement, the counts in the first measurement, and the time measurement

$$T = \frac{\ln(2)}{\ln(N_0) - \ln(N_i)} t_i \tag{16}$$

so

$$\delta T = \sqrt{\left(\frac{\partial T}{\partial N_i} \delta N_i \right)^2 + \left(\frac{\partial T}{\partial N_0} \delta N_0 \right)^2 + \left(\frac{\partial T}{\partial t_i} \delta t_i \right)^2} \tag{17}$$

throwing this into python again gives

$$\langle \sigma \rangle = 0.01 \tag{18}$$

i.e

$$T = 1.04 \pm 0.01\text{h} \tag{19}$$

and the chi-square is

$$\chi^2 = 145 \tag{20}$$

which is very high, meaning something is not right.

4 Scholarship winners

4.1 Assuming Poisson

If we assume the distribution is poisson for each country, then the parameter λ for each is

$$\lambda_i = n_i p_i = n_i \left(\frac{winners}{1 \times 10^5} \right) = winners \cdot \frac{n_i}{1 \times 10^5} \tag{21}$$

and the standard deviation is the square root

$$\sigma_i = \sqrt{\lambda_i} \tag{22}$$

plugging in the numbers the table becomes

Country	Population	winners per 100k	error
Israel	9×10^6	3.6	0.2
Germany	82.8×10^6	4.0	0.07
Denmark	5.8×10^6	3.5	0.25
USA	325.7×10^6	10.9	0.06
Sri Lanka	105×10^3	1.9	1.35
Ivory Coast	33×10^3	3.2	3.11
Puerto Rito	22×10^3	4.8	4.8

4.2 Assuming Gaussian

The number of winners per 100k in Israel is 3.6 and the error is 0.2. The number of winners per 100k in Germany is 4.0. Since we assume the distribution is gaussian, we can look at the difference between Germany and Israel and express it in standard deviations of the Israeli count

$$4 - 3.6 = 0.4 = 2\sigma_{Israel} \tag{23}$$

So the chance the number of winners in Israel surpasses that of Germany is

$$1 - \Phi(2) = 1 - 0.9772 = 0.0228 = 2.28\% \tag{24}$$

5 Measurements with multiple categories

5.1 Estimators

First the linear fit is

$$x = x_0 + vt \tag{25}$$

using the formulas from training 6

$$\hat{v} = \frac{\overline{tx} - \overline{t}\overline{x}}{\overline{t^2} - \overline{t}^2} \qquad \hat{x_0} = \overline{x} - \hat{v}\overline{t} \tag{26}$$

and

$$\sigma_{\hat{v}}^2 = \frac{\sigma^2}{N\left(\overline{t^2} - \overline{t}^2\right)} \qquad \sigma_{\hat{x_0}}^2 = \frac{\sigma^2\overline{t^2}}{N\left(\overline{t^2} - \overline{t}^2\right)} \tag{27}$$

calculating using python

Cart	velocity estimate (cm/s)	estimate error
1	0.280	0.095
2	0.280	0.095
3	0.280	0.095
4	0.253	0.095

5.2 chi square

by definition

$$\chi^2 = \frac{1}{\sigma^2}\sum_{i=1}^{N}(x_i - x_0 - vt_i)^2 \tag{28}$$

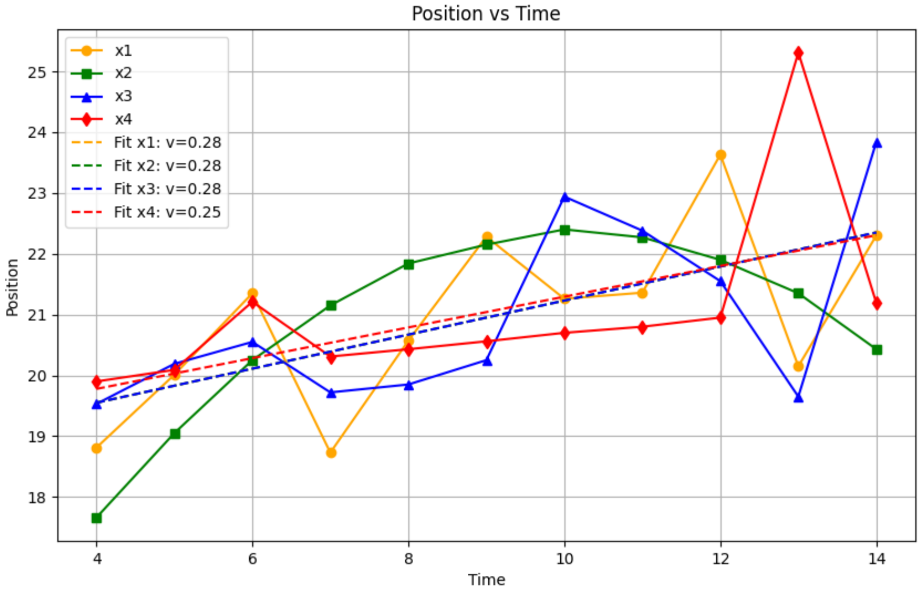
plugging the numbers in using python

Cart	chi-square
1	13.75
2	13.75
3	13.75
4	14.77

the chi-square is higher than the degrees of freedom (9, from 11 measurements minus 2 parameters) for all the carts, so none of the measurements appear to be indicative of a constant velocity.

5.3 Plot

the graph is



and indeed none of them seem to be moving at a constant velocity (or anywhere near constant)